

FPGA-Accelerated ECG Filtering and Peak Analysis

Experiment 3 (Intermediate) | V-SPACE FPGA Build Challenge 2026

Field	Value
Platform	PYNQ-Z2 / Zynq XC7Z020 (xc7z020clg400-1)
Tools	Vitis HLS / Vivado 2025.1.1; PYNQ Python; independent Icarus Verilog 12.0 RTL verification
Confirmed members	Shanshank Pulipati - 25BEC0573; Kartik Yadav - 25BEC0087
Team name	Beyond Boolean

1. Objective

Accelerate 128-tap, 16-bit Q4.12 FIR preprocessing in FPGA programmable logic, transport samples through AXI DMA and verify the numerical output against an independent fixed-point reference. Demonstrate candidate ECG peak detection and heart-rate estimation in Python on the ARM processor using recorded MIT-BIH data. Live ECG sensor acquisition and clinical arrhythmia classification are outside the demonstrated scope.

2. Block Diagram

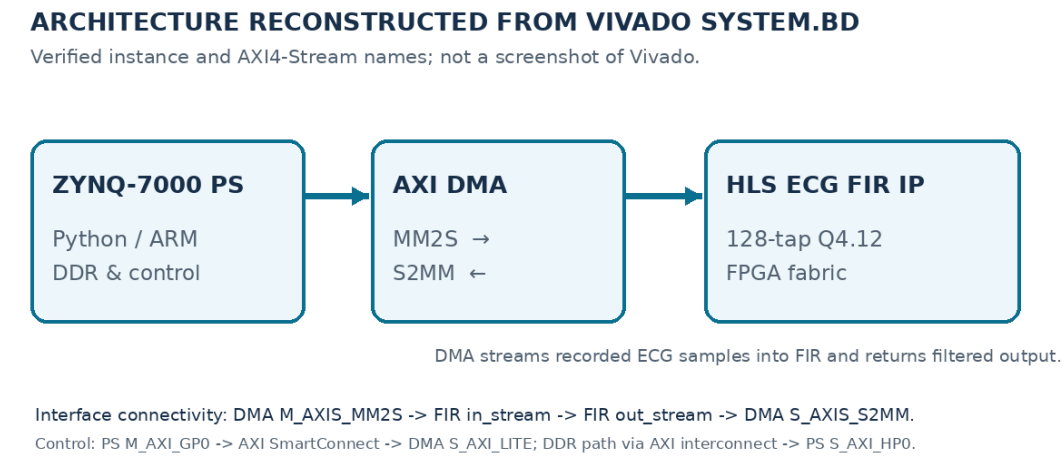


Figure 1. Architecture reconstructed from verified system.bd net names. This is a source-derived block diagram, not a Vivado GUI capture.

DDR samples pass through AXI DMA MM2S, the HLS FIR in_stream/out_stream interfaces and DMA S2MM back to memory. The Zynq PS controls DMA over AXI-Lite and accesses memory through the high-performance interconnect path.

3. RTL Design and Module Hierarchy

Module / source	Responsibility
ecg_bandpass_filter.cpp / .h	Original HLS C++ FIR function, fixed-point types and stream interface
ecg_bandpass_filter.v	Actual generated FIR top, count=2048 in the board design, active-low control reset
Pipeline_VITIS_LOOP_79_1 and mac_muladd modules	Generated pipelined filtering and multiply-add dependencies
regslice_both	Generated stream register slices
system_wrapper.v / system.bd	Vivado PS, DMA, interconnect and FIR integration

The expanded original IP package and all generated Verilog dependencies are supplied. The Vivado wrapper requires its block-design/vendor-IP hierarchy; the full original workspaces remain linked in SOURCE_PROVENANCE.md. Supplied XDC files are generated IP/out-of-context constraints rather than a claimed custom board pin constraint file.

4. Simulation Results

Original Vitis HLS C-simulation logs show a pass. Independent Icarus Verilog 12.0 simulation on 9 October 2026 now additionally checks the unchanged generated RTL. Four stimulus groups include impulse, sine, seeded signed random values and recorded ECG. Input gaps, output stalls, repeated 2048-sample batches, KEEP/STRB preservation and TLAST are tested.

Verification	Observed result
Independent RTL numerical check	28,672/28,672 exact samples; maximum absolute error 0 LSB
Stimulus composition	20,480 payload + 8,192 zero-flush samples
Batches / TLAST	14 batches; all frame and invocation counts pass
Stream checks	Output stability during backpressure; KEEP/STRB and TLAST pass
Reference cross-check	ECG integer reference agrees with saved physical-board output CSV

The generated FIR static state is cleared between independent inputs with a full zero batch, matching the existing board notebook. The testbench waits for ap_done and pipeline state writeback. The report does not assume that control reset alone clears FIR history.

3. RTL Design - Actual Vivado Capture

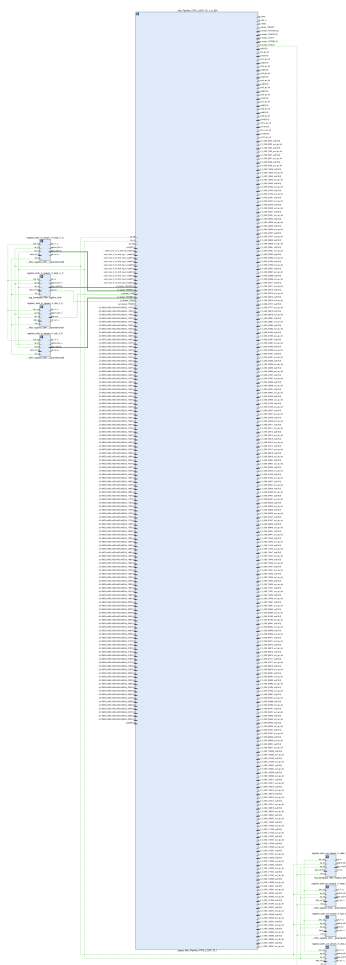


Figure 2. Actual Vivado 2025.1.1 elaborated FIR hierarchy exported by the team: one generated filtering pipeline and eight AXI-stream register-slice instances (9 cells, 28 displayed nets). The full vector PDF is supplied as *Images/rtl_schematic.pdf* for detailed zooming. This is a selected module hierarchy view; top-level primitive logic is retained in the source files.

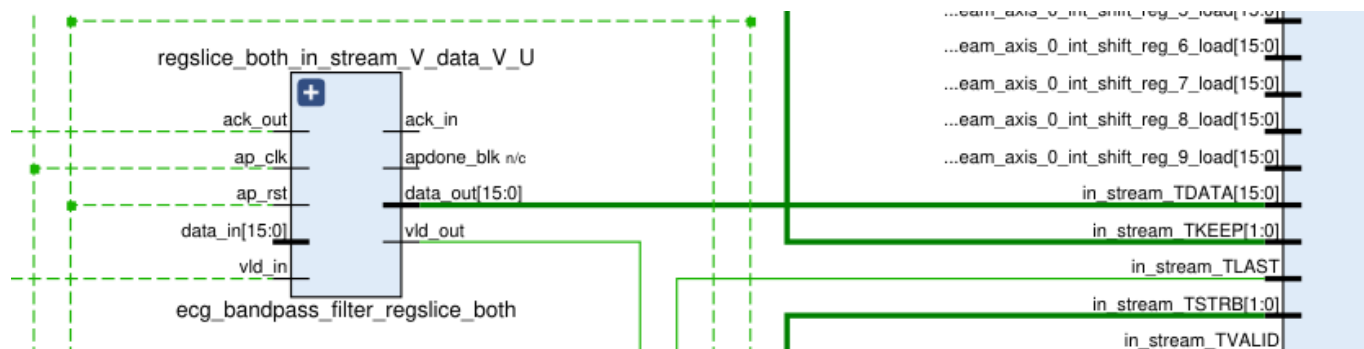


Figure 3. Magnified input-register-slice excerpt rendered directly from the same unmodified Vivado PDF. Labels and connections are preserved.

4. Simulation Results - Actual Waveform

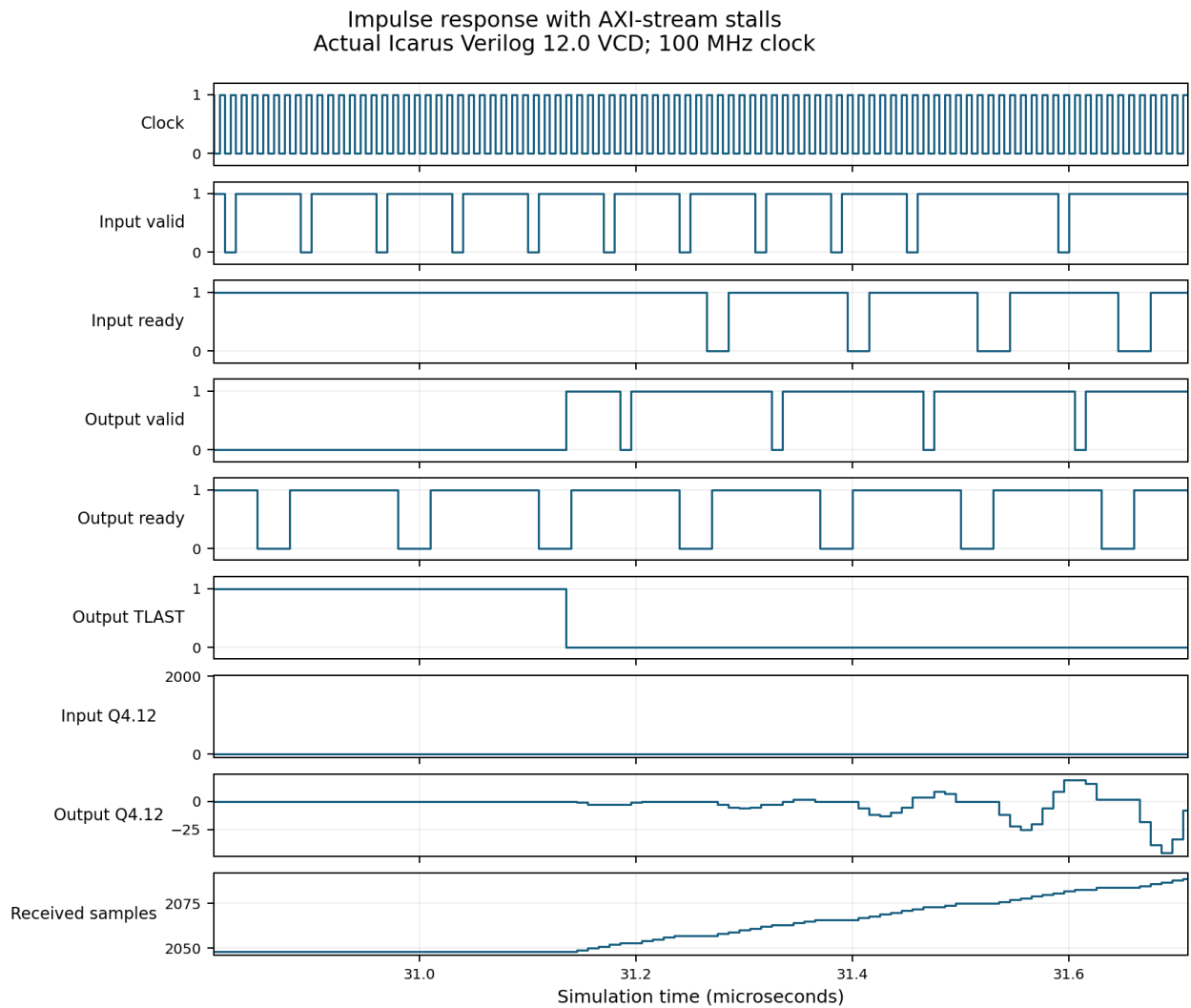


Figure 4. Actual generated-RTL waveform decoded from the saved Icarus VCD. All waveform points come from the executed simulation; this is not a conceptual timing drawing.

The full run is preserved in `Simulation/rtl_waveform.vcd.gz`. `Simulation/transcript.txt` contains the commands and actual passing outputs; `rtl_output_samples.csv` contains all 28,672 checked samples. The standalone simulation report provides the test scope and acceptance criteria. This unit simulation does not model the entire Zynq PS/DDR/DMA system.

5. Hardware Implementation

The original physical PYNQ-Z2 demonstration loaded the overlay and detected axi_dma_0. The matching BIT and HWH are retained unchanged. The XSA contains the same bitstream. The screenshot below is a recovered original Jupyter capture showing the loaded overlay and available IP; it is not a fresh board run.

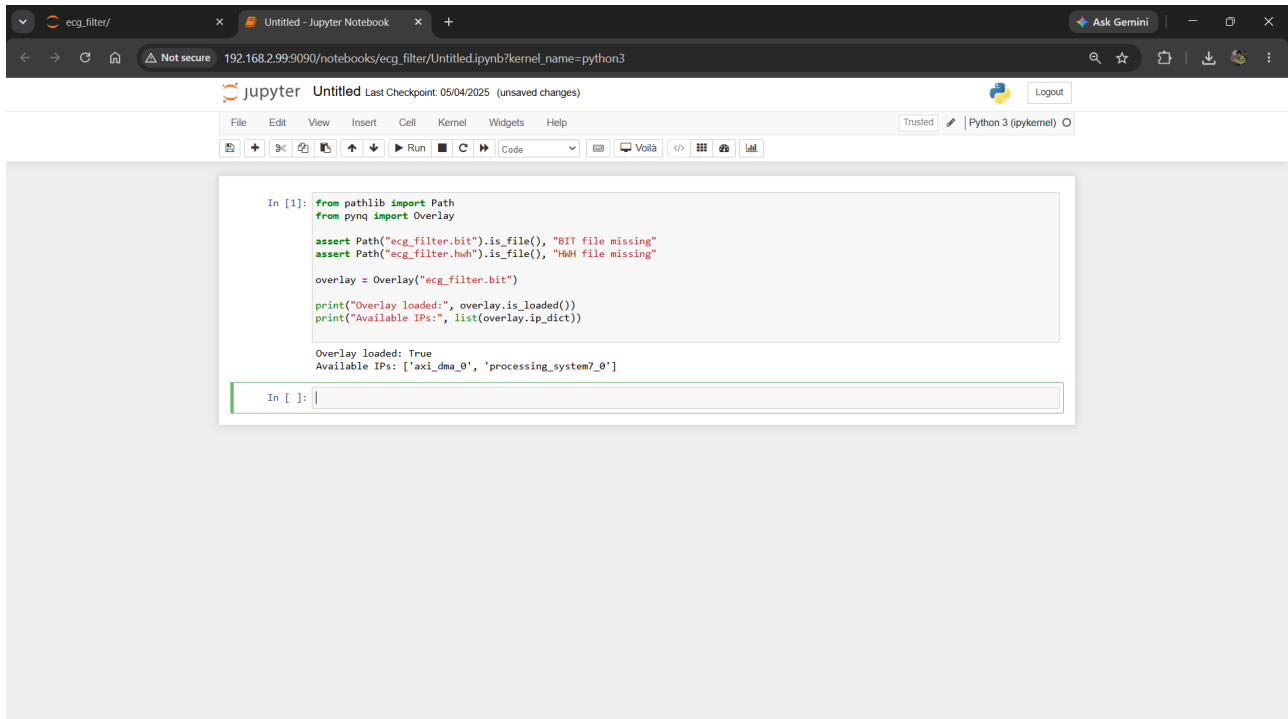


Figure 5. Original Jupyter screenshot: Overlay loaded: True; axi_dma_0 and processing_system7_0 detected. The original PNG is preserved; hardware_output.jpg is its uncropped format derivative.

Saved physical-board measurement	Result
Synthetic fixed-point comparison	2,048/2,048 exact samples; 0 Q4.12 LSB maximum error
5 Hz gain / 50 Hz reduction	0.9940 / 53.74 dB in the demonstrated sine test
Recorded ECG comparison	12,288/12,288 exact; six 2048-sample data batches
Python candidate peak analysis	42 peaks; mean RR about 0.812 s; heart rate 73.9 bpm
Python + DMA wall time	7.55 ms in the saved run; not FPGA-only latency

Whole-project placed resources from the original Vivado report: 7,232 slice LUTs, 13,844 slice registers, 2 block RAM tiles and 40 DSPs. These are not FIR-only counts. The original timing and routed DRC reports remain supplied.

5. Hardware Implementation - Saved Output Plots

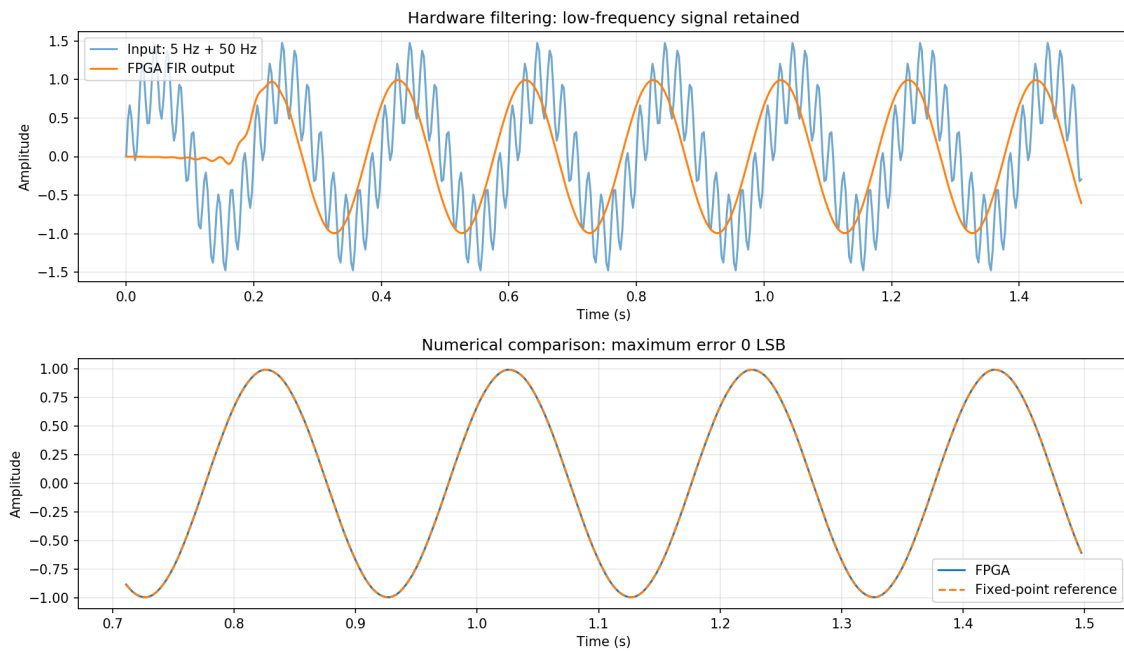


Figure 6. Original saved board sine-response and fixed-point comparison plot.

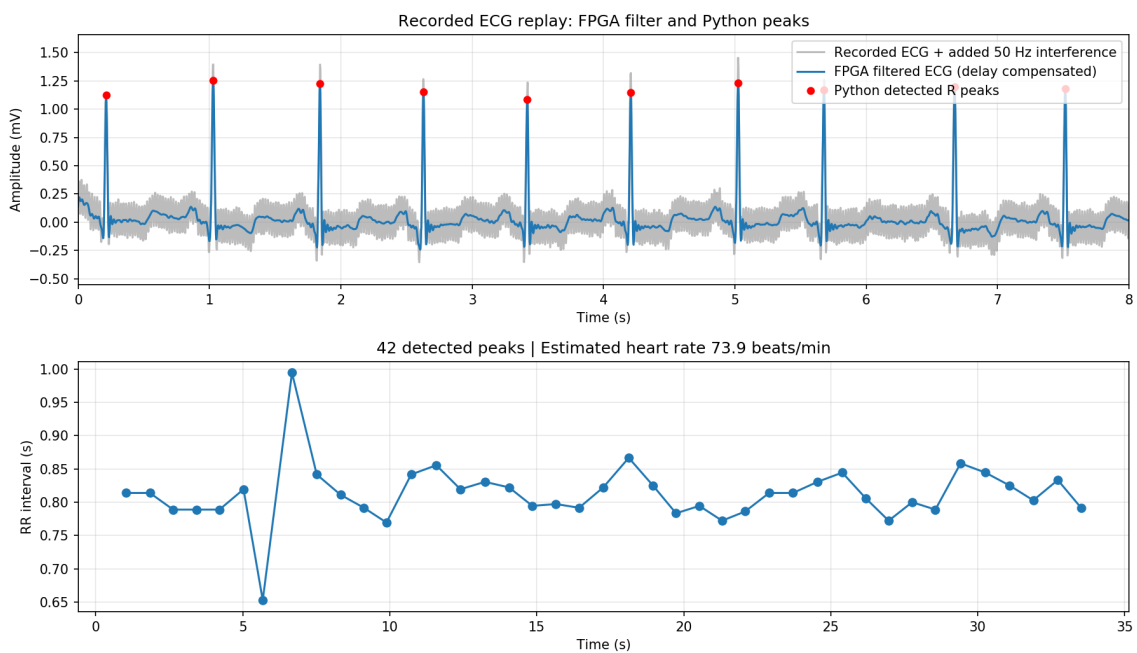


Figure 7. Original saved board ECG replay and Python peak analysis. The record is MIT-BIH 100 MLII, sampled at 360 Hz, with 0.15 mV added 50 Hz interference. FIR filtering is FPGA logic; peak detection is Python on ARM.

The actual setup photograph and Vivado capture have now been supplied and are incorporated in this report.

5. Hardware Implementation - Physical Setup



Figure 8. Actual team-supplied PYNQ-Z2 setup photograph, supplied 9 October 2026. The USB and Ethernet connections and illuminated board LEDs are visible. The original JPEG bytes are retained as `Images/board_setup.jpg`. The photo documents the setup; numerical hardware-test evidence remains in the saved executed notebooks and result files.

6. Applications

Educational PS/PL biomedical signal processing; FPGA preprocessing for embedded signal-analysis systems; and demonstrations of fixed-point accuracy, memory-to-stream DMA and hardware/software co-design. Medical diagnosis would need separate sensor interfacing and validation; that capability is not claimed by this demonstration.

7. Conclusion and Future Scope

The saved physical-board demonstration verifies exact filtering for synthetic and recorded ECG inputs and performs candidate peak/heart-rate analysis on ARM. The additional independent RTL test verifies exact core samples and tested stream protocol behavior under stalls and across batch boundaries. Future work includes live acquisition, broader signal validation, hardware peak detection and controlled performance benchmarking.

Evidence and Reproduction

Original HLS/Vivado workspaces, the original packaged IP and saved physical-board results are preserved or linked in SOURCE_PROVENANCE.md. Testbench/run_rtl_sim.py reproduces the independent unit simulation. The waveform PNGs are derived from its actual VCD. rtl_run_metadata.json identifies the simulator and hashes the tested RTL/testbench files. FILE_MANIFEST.json checksums the package.

MIT-BIH Arrhythmia Database v1.0.0, record 100 MLII, samples 0-12287, 360 Hz. Source: <https://physionet.org/content/mitdb/1.0.0/>. Attribution/license details are in PYNQ_Hardware/DATA_SOURCE.md.

Submission Status

The Project 3 file set now includes the required four images, Verilog testbench, actual RTL waveform/transcript/simulation report, confirmed team details and this seven-section project report. Team: Beyond Boolean; Shanshank Pulipati (25BEC0573) and Kartik Yadav (25BEC0087). The final demo URL is recorded in Video_Link.txt; see Documentation/VIDEO_REVIEW.md for content and access review.

The seven-section format follows the organizer-provided documents_req.pdf. The full five-project report is a separate team-level deliverable.